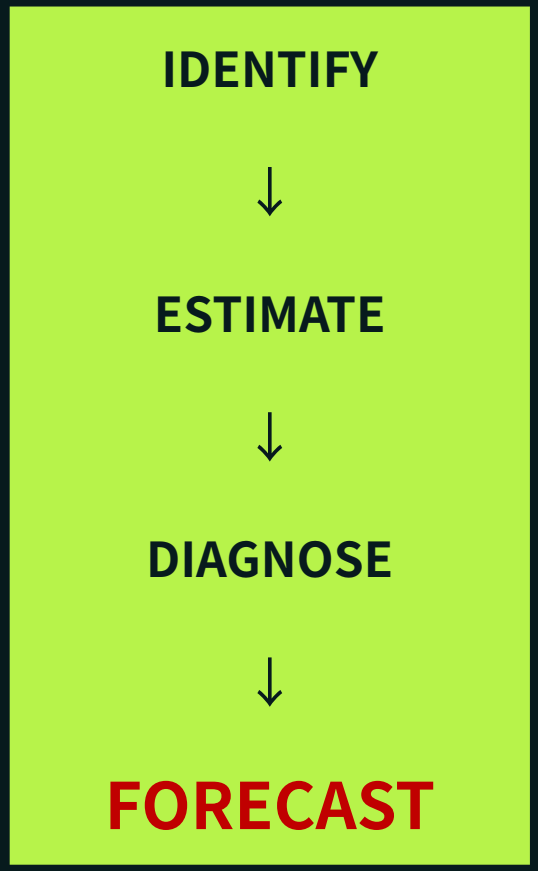
The Box–Jenkins Procedure

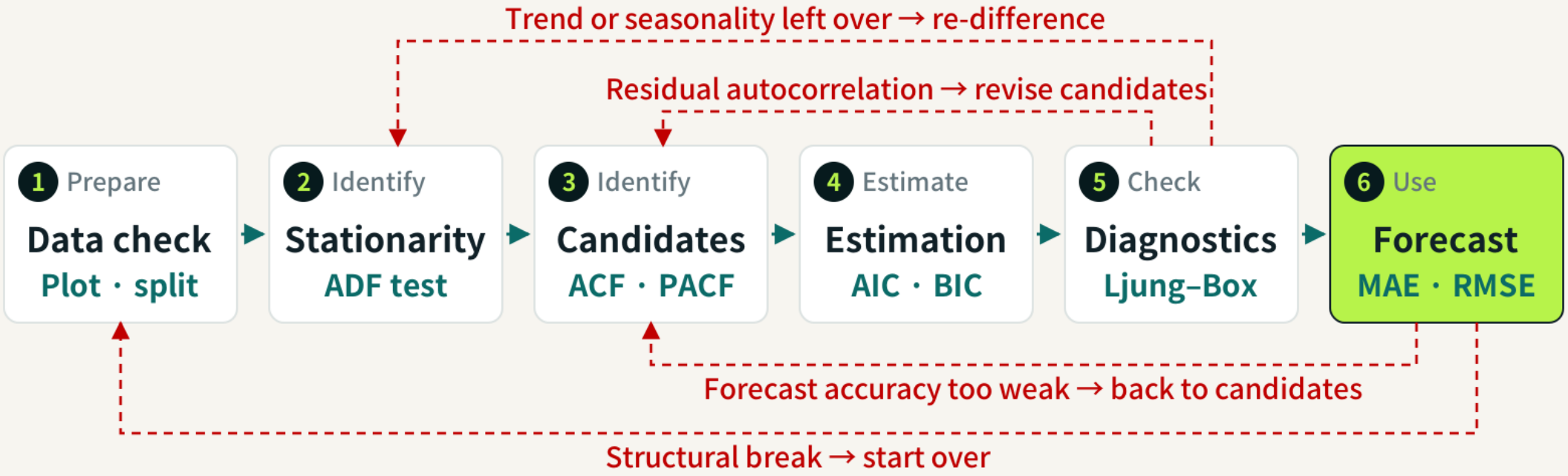
A six-step cheat sheet for ARIMA forecasting



Time Series Forecasting · ARIMA Modelling

Six steps run forward, but diagnostics send you back

Box & Jenkins (1970): identification → estimation → diagnostic checking → forecasting



ADF: difference? · AIC / BIC: which model? · Ljung–Box: anything left?

Every red arrow is a normal part of the procedure, not a failure.

ARIMA(p,d,q) combines past values, differencing and past shocks

The model family the procedure identifies; B is the backshift operator

$$\underbrace{\phi(B)}_{\mathrm{AR}(p)}\ \underbrace{(1-B)^d}_{\mathrm{I}(d)}\ y_t = \underbrace{\theta(B)}_{\mathrm{MA}(q)}\ \varepsilon_t$$

$$B\,y_t = y_{t-1}$$
$$\phi(B) = 1 - \phi_1 B - \cdots - \phi_p B^p$$
$$\theta(B) = 1 + \theta_1 B + \cdots + \theta_q B^q$$

AR(p) · AUTOREGRESSIVE

Past values

p lagged values of the series itself

I(d) · INTEGRATED

Differencing

d rounds of differencing to remove the trend

MA(q) · MOVING AVERAGE

Past shocks

q past errors — not a rolling mean of the data

The six steps are a disciplined way to choose p, d and q — and to check the choice.

ARMA needs a stationary series, so differencing comes first

Stationary = statistical properties that do not drift over time

NON-STATIONARY

- Trend or random walk
- Seasonal unit root
- Shocks never fade

difference →

STATIONARY

- Constant mean and variance
- Correlation set by lag only
- ACF / PACF readable

Regular difference · seasonal difference (s = 12 for monthly data)

$$\Delta y_t = y_t - y_{t-1} \qquad \Delta_s y_t = y_t - y_{t-s}$$

Difference only as much as needed — over-differencing adds noise.

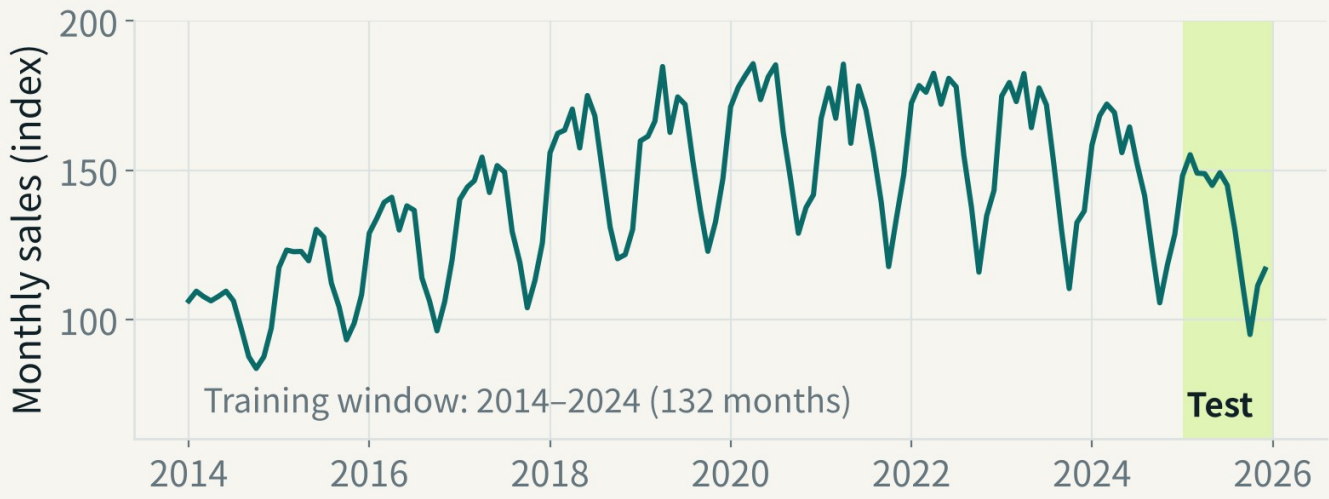
Step 1 · Plot the series before modelling anything

Simulated monthly sales, Jan 2014 – Dec 2025 (144 months)

CHECKLIST

- Target, frequency, horizon
- Time order, gaps, outliers, units
- Trend, seasonality, widening swings

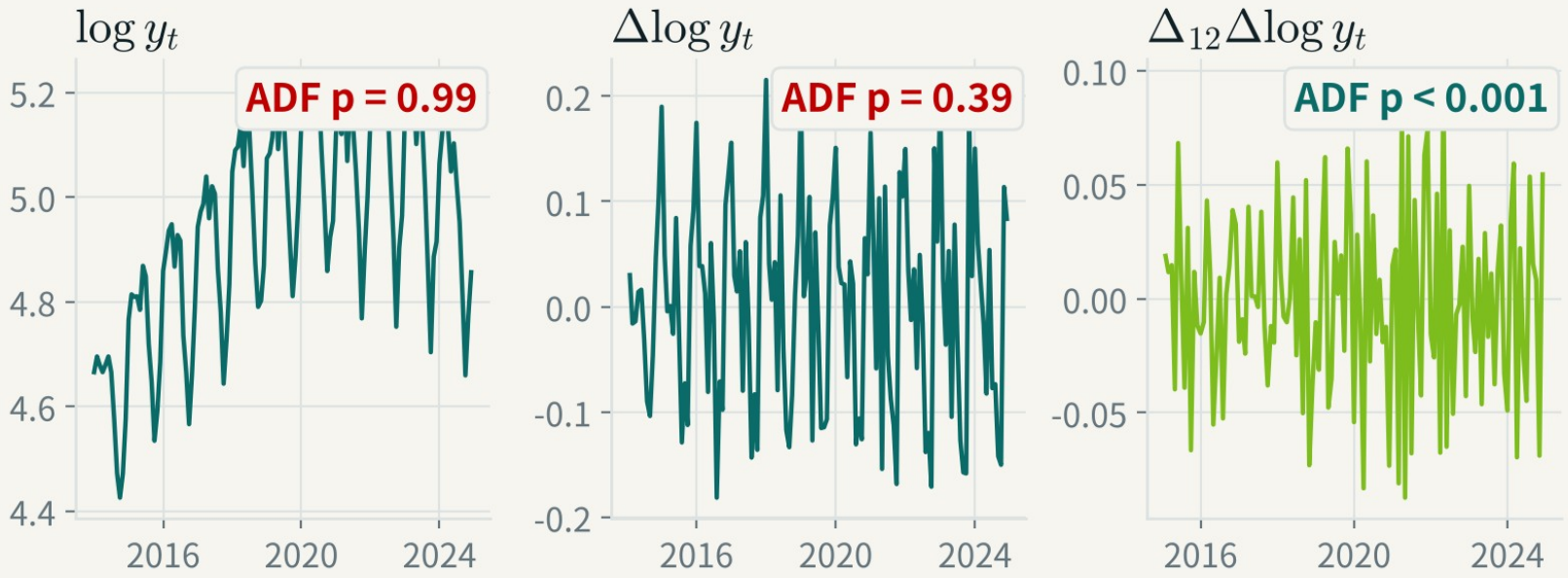
Output: clean series + train / test split



Split by time, never at random — the last 12 months stay untouched until Step 6.

Step 2 · Log and two differences make the series stationary

ADF p-values on the 132-month training window; s = 12 for monthly data



Level p = 0.99; first difference p = 0.39 — the seasonal pattern still blocks stationarity.

Only the seasonal difference on top passes: set d = 1, D = 1.

Step 2 · The ADF test asks whether a unit root is present

Augmented Dickey–Fuller regression; lag length k chosen by AIC

ADF regression

$$\Delta y_t = \alpha + \beta t + \gamma\, y_{t-1} + \sum_{j=1}^{k} \delta_j\, \Delta y_{t-j} + \varepsilon_t$$

Hypotheses

$$H_0 :\ \gamma = 0 \quad (\text{unit root})$$
$$H_1 :\ \gamma < 0 \quad (\text{stationary})$$

p < 0.05

Reject the unit root

Confirm with the plot, then Step 3

p ≥ 0.05

Cannot reject

Difference again and repeat Step 2

※ Match constant/trend settings; standard ADF does not test seasonal unit roots.

ADF is a clue, not a verdict: difference only as much as needed.

Step 3 · Cut-off versus slow decay separates AR from MA

Theoretical signatures of the ACF and PACF; real samples are noisier

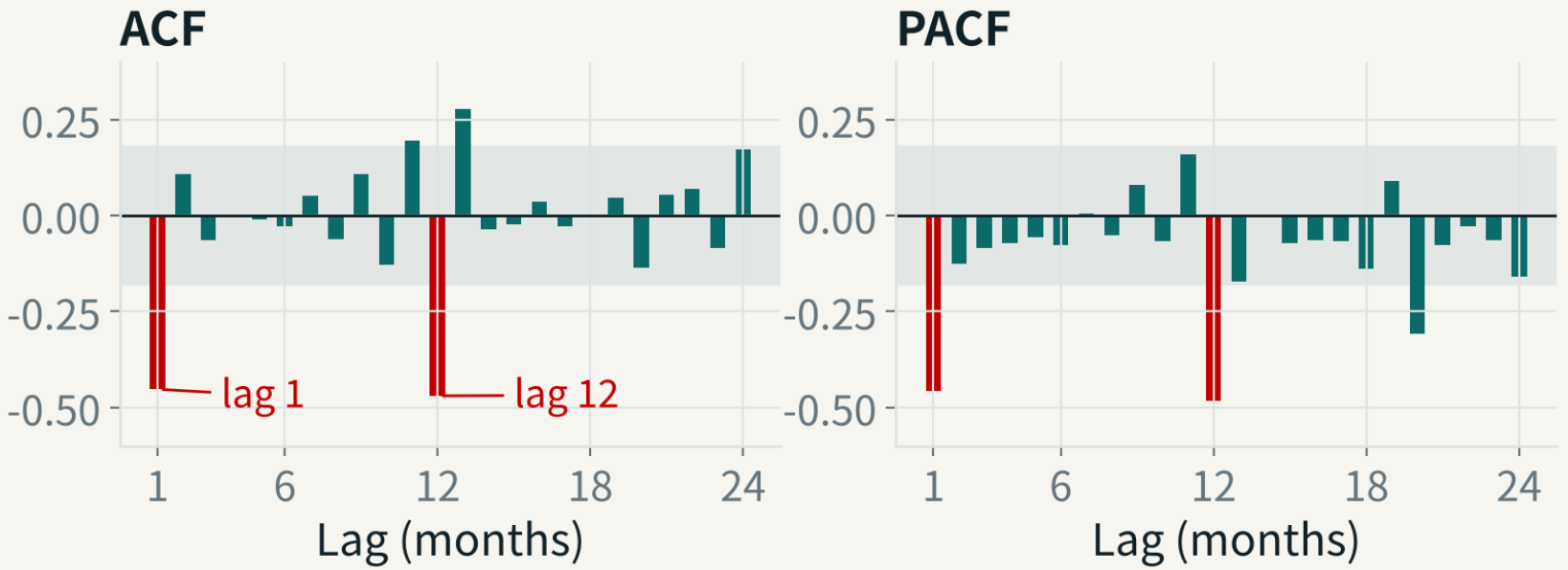
ACF	PACF	Candidate to try first
Decays slowly	Cuts off after lag p	AR(p)
Cuts off after lag q	Decays slowly	MA(q)
Decays slowly	Decays slowly	ARMA(p,q)
Spikes at lags s, 2s, …	Spikes at lags s, 2s, …	Seasonal AR / MA

※ ACF: correlation between today and lag j. PACF: the same correlation after controlling for lags 1 … j−1.

Use the rules to shortlist small models — never to pick one final model.

Step 3 · Our series shows MA spikes at lag 1 and lag 12

ACF and PACF of the seasonally and regularly differenced log series; grey band = 95% limits



ACF: -0.45 at lag 1 and -0.47 at lag 12, then cuts off; PACF decays around both.

Shortlist: MA(1) with a seasonal MA(1), plus three close neighbours.

Step 4 · AIC and BIC reward fit and penalise extra parameters

Estimate every shortlisted candidate by maximum likelihood, then compare

Akaike information criterion

$$\mathrm{AIC} = -2\log\hat{L} + 2k$$

Bayesian information criterion

$$\mathrm{BIC} = -2\log\hat{L} + k\log n$$

FIT

Higher likelihood

Better fit lowers both criteria

PENALTY

Parameters k

BIC's penalty also grows with n

DECISION RULE

Lower is better

Disagree? Keep both for Steps 5–6

※ **Compare only models fitted to the same data, transform and differencing.**

AIC and BIC are selection criteria, not hypothesis tests.

Step 4 · The airline model wins on both AIC and BIC

Seasonal ARIMA (0,1,1)(0,1,1) with s = 12 on log sales — the classic “airline model”

$$(1 - B)(1 - B^{12}) \log y_t = (1 + \theta_1 B)(1 + \Theta_1 B^{12})\, \varepsilon_t$$

Candidate $(p,d,q)(P,D,Q)_{12}$	Parameters k	AIC	BIC
$(0,1,1)(0,1,1)_{12}$	**3**	**-477.9**	**-469.6**
$(0,1,1)(1,1,0)_{12}$	3	-476.7	-468.4
$(1,1,1)(0,1,1)_{12}$	4	-476.5	-465.4
$(1,1,0)(0,1,1)_{12}$	3	-475.8	-467.5

Lowest on both criteria — check convergence and invertibility, then diagnose.

Step 5 · Residuals should look like white noise

Residual = actual value minus the one-step-ahead forecast

Residual

$$\hat{e}_t = y_t - \hat{y}_{t|t-1}$$

Ljung–Box test

$$Q = n(n+2)\sum_{j=1}^{h}\frac{\hat{\rho}_j^2}{n-j} \sim \chi^2_{h-m}$$

$$H_0: \ \rho_1 = \cdots = \rho_h = 0, \quad m = \text{no. of AR/MA terms}$$

p < 0.05

Autocorrelation left → Step 3

Check the seasonal lags first

p ≥ 0.05

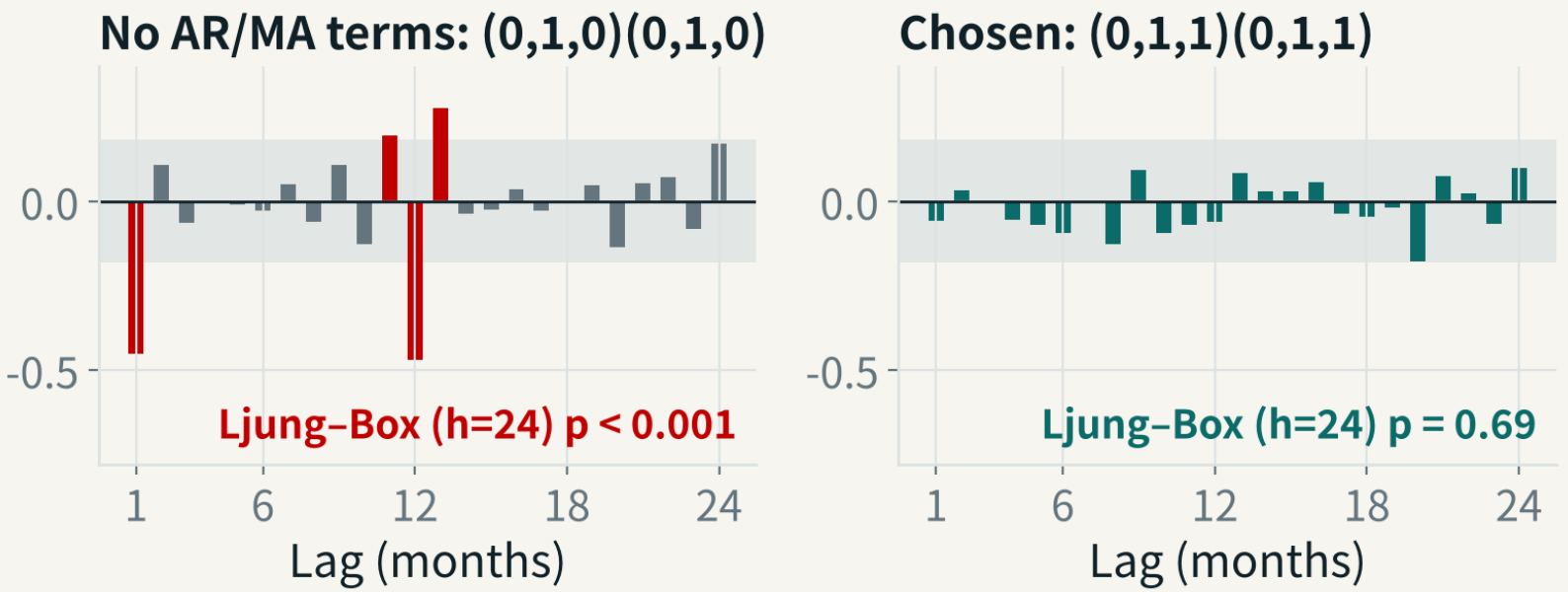
No clear evidence → Step 6

Still check variance shifts and breaks

A large p-value does not prove independence, normality or a perfect model.

Step 5 · The MA terms absorb the autocorrelation left behind

Residual ACF of a model without AR/MA terms versus the chosen airline model

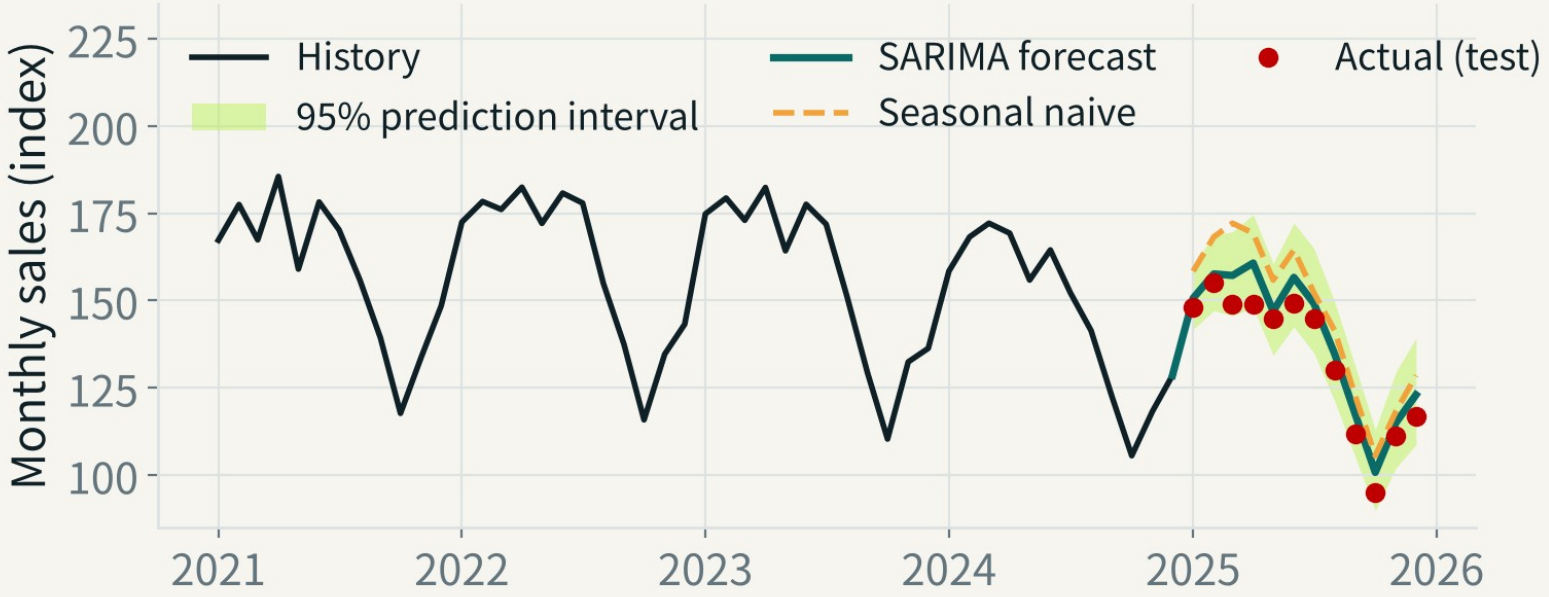


Without MA terms, lags 1 and 12 stay far outside the band and Ljung–Box rejects.

Chosen model: Ljung–Box p = 0.69 at h = 24 — no clear pattern left.

Step 6 · The model beats the seasonal naive forecast on unseen data

12-month forecast from Dec 2024 versus the held-out test year



All 12 test months fall inside the 95% interval; MAE 5.4 vs 12.7 for seasonal naive.

Always compare against a simple benchmark before trusting a model.

Step 6 · Score forecasts on the test window, then keep updating

H = 12 forecast months from origin T; lower is better for both measures

Mean absolute error

$$\mathrm{MAE} = \frac{1}{H}\sum_{h=1}^{H}\left|y_{T+h} - \hat{y}_{T+h|T}\right|$$

Root mean squared error

$$\mathrm{RMSE} = \sqrt{\frac{1}{H}\sum_{h=1}^{H}\left(y_{T+h} - \hat{y}_{T+h|T}\right)^2}$$

Method	MAE	RMSE	Error vs naive
SARIMA $(0,1,1)(0,1,1)_{12}$	**5.4**	**6.1**	**–57%**
Seasonal naive	**12.7**	**13.5**	**—**

※ Tuned on the test window? Hold out a separate final test period.

Refit on all data and report forecast plus interval; a break sends you back to Step 1.

Three tools, three questions, one loop

Each tool supports one decision; none of them replaces looking at the plots

STEP 2 · ADF

Do I need to difference?

A clue for d and D — read together with the plot

STEP 4 · AIC · BIC

Which candidate is best?

Compare models with the same data and differencing

STEP 5 · LJUNG–BOX

Is anything left in the residuals?

Rejection sends you back to Step 3

Plot → stationarise → shortlist → estimate → diagnose → forecast, and loop when needed

Identify with the ACF and PACF.

Choose with AIC and BIC.

Trust only what the residuals and the test year confirm.

The Box–Jenkins Procedure
A Six-Step Cheat Sheet

※ All figures use a simulated monthly series (seed 7) built from an airline-type process; statsmodels 0.14.